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THE NATURAL EXCITATION TECHNIQUE (NExT) FOR MODAL PARAMETER EXTRACTION FROM OPERATING STRUCTURES

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ABSTRACT

The Natural Excitation Technique (NExT) is a method of modal testing that allows structures to be tested in their ambient environments. This paper contains developments and results since 1990, and a new theoretical derivation of NExT, as well as a verification using analytically generated data. In addition, results from NExT are compared with conventional modal testing for a parked, vertical-axis wind turbine. Also, for a rotating turbine, NExT is used to calculate the modal parameters as functions of the rotation speed, since substantial damping is derived from the aeroelastic interactions during operation. Finally, experimental results calculated using NExT are compared with analytical predictions of damping using aeroelastic theory.

1. Introduction

The basis for extracting modal parameters from structures undergoing ambient excitation has a long history dating to the 1960s. There have been several different approaches to the analysis of such data: peak-picking from Power Spectral Density (PSD) functions, Auto Regressive–Moving Average (ARMA) models, the Maximum Entropy Method (MEM), random decrement processing coupled with time-domain parameter extraction, direct use of raw data with time-domain parameter extraction, and the use of cross-correlation functions with time-domain parameter extraction. Akaike was among the first to study the use of Auto Regressive–Moving Average (ARMA) type models to analyze systems with ambient excitation [1]. This procedure was applied to a variety of systems including machine tools [2]. The closely associated Maximum Entropy Method (MEM) has been applied to the analysis of offshore structures undergoing wave excitation [3]. Another procedure called random decrement [4] was developed to process ambient-excitation data. This processing allowed the first use a time-domain parameter extraction scheme, specifically the Ibrahim Time Domain (ITD) technique [5], to be applied ambient-excitation data [6]. Later work proved that the random decrement technique was an alternative method for estimating the auto-correlation function [7]. Pappa and Juang later utilized naturally-excited data directly in the Eigensystem Realization Algorithm (ERA) [8] which is also a time-domain parameter extraction scheme [9]. It has also been proposed that the use of data directly in the ERA with Data Correlations (ERA/DC) algorithm [10] is equivalent to using cross-correlation functions in the original ERA algorithm [11].

Clarkson and Mercer presented a procedure in 1965 to estimate the frequency response characteristics of a structure excited by white noise using cross-correlation functions [12]. Although their work assumed a measured input, the conceptual framework was laid to utilize correlation functions instead

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of impulse response functions when the excitation was not known. When coupled with a time-domain modal extraction scheme [5, 8, 13, 14], this concept became a very powerful tool for the analysis of naturally excited structures. This concept was applied by James et al. [15] to wind turbines and Kim et al. [11] to a launch vehicle.

The concept of using natural excitation for modal testing of parked Vertical-axis Wind Turbines (VAWTs) was first suggested by Lauffer et al. [16]. The technique was developed further and used for the modal tests of other VAWTs [15, 17]. Upon the development of a theoretical basis for utilizing cross-correlation functions in time-domain parameter extraction algorithms and several validation exercises [18], the term Natural Excitation Technique (NExT) was used to refer to this formalized approach. NExT is a method of modal testing that allows structures to be tested in their ambient environments. Initially, this procedure was developed to allow VAWTs to be tested during parked and operating conditions. Since, these structures are much different in the operating condition as compared to the parked condition, the modal frequencies and damping ratios must be extracted from measured response data. More specifically, knowledge of the total modal damping (structural and aeroelastic) is important for predicting fatigue life and reducing resonant responses. NExT has also been applied to the verification of aeroelastic damping models for rotating VAWTs [18–20], a three-bladed composite VAWT [21], a Horizontal-axis Wind Turbine (HAWT) [22], a rocket during launch [23], a ground transportation vehicle [18], a bridge [24], and an offshore structure [21].

A brief overview of NExT is presented, followed by its theoretical development and a demonstration using analytically generated data. Modal tests using step relaxation and wind excitation (NExT) are compared for a parked 19-m turbine manufactured by FloWind Corporation. The results show the accuracy of NExT and uncover a damping mechanism for parked wind turbines. A rotating wind turbine, which derives substantial damping from aeroelastic interactions during operation, was tested and the modal parameters were determined as a function of the operating rotation speed using NExT. This provides an experimental technique to quantify the aeroelastic properties of wind turbines. Further, experimental results calculated using NExT are compared with analytical predictions of damping using aeroelastic theory. These measurements provide new information to refine aeroelastic theories that predict damping.

2. Overview

Conventional modal analysis utilizes frequency response functions (FRFs) or impulse response functions which require measurements of both the input force and the resulting response. However, ambient excitation (such as wind, wave, road noise, and traffic excitation) does not lend itself to FRF calculations because the input force cannot be measured. NExT is a four-step process designed to estimate modal parameters of structures excited in their operating environment. The first step is to acquire response data from the operating structure. Sensors that can measure strain, displacement, velocity, or acceleration response are required. Long time histories of continuous data are desired to allow significant averaging of the data, provided the operating conditions are relatively stationary.

The second step is to calculate auto- and cross-correlation functions from these time histories using standard techniques [25, 26]. Correlation functions are commonly used to analyze randomly excited systems [26]. As the following section will show, the correlation functions can be expressed as summations of decaying sinusoids. Each decaying sinusoid has a damped natural frequency and damping ratio that is identical to that of a corresponding structural mode. The cross-correlation of each output channel will be calculated with respect to a subset of the output channels which function as references. These output-references are chosen to contain all of the relevant modal information and/or to target specific structural responses. In this manner, selection of these output-reference channels is similar to choosing the input-reference locations in a traditional modal test. An appropriate choice can aid in the separation of closely spaced modes and the proper extraction of low-amplitude modes. Multiple output-references were used in each of the examples presented in this work.

The third step of NExT uses a time-domain modal identification scheme to estimate the modal parameters by treating the correlation functions as though they were free vibration responses—that is, sums of decaying sinusoids. The Polyreference technique [14] and the Eigensystem Realization Algorithm (ERA) [8] have been used as the time-domain modal identification schemes in this work to extract modal frequencies and damping ratios. However, for single-input systems, Ibrahim Time Domain (ITD) [5] or Least Squares Complex Exponential [13] could be used.

The final step of NExT extracts mode shape information using the identified modal frequencies and modal damping ratios. Work has been performed on shape extraction [17, 22]; however, this report will not present any shape results. It should be noted that it will not be possible to calculate the mass-scaling for the mode shapes. This calculation requires a measurement of the input force which is not possible with natural excitation. An activity closely related to shape extraction uses the identified modal parameters to synthesize the autospectrum from each sensor. This provides a means of visually verifying the accuracy of the estimated modal frequencies and damping ratios. This paper will present results from autospectrum synthesis.

3. Theoretical Development of NExT

A theoretical justification of NExT entails proving that a MIMO (multiple input, multiple output), multiple-mode system excited by random inputs produces autocorrelation and cross-correlation functions that are sums of decaying sinusoids. Furthermore, these decaying sinusoids must have the same frequencies and damping ratios as the structural modes. Consequently, the correlation functions will have the same form as impulse response functions and can be used in standard modal analysis algorithms.

The approach is to develop a general solution for a structure with a discrete spatial representation; define the cross-correlation function between two outputs; and solve for the case of random inputs. The theoretical justification of NExT can be developed for a general class of random inputs, fully complex modes, and the presence of known harmonic inputs. However, this development will be limited to the case of white-noise inputs, real modes, and no harmonics, thus allowing the reader to obtain an appreciation for the theoretical background of NExT without the added complexities of the most general case. The derived form of the cross-correlation function will then be simplified to provide the reader with a form very similar to that of the impulse response function. Then verifications of the derived result will be performed using a standard cross-correlation identity and an autocorrelation identity.

4. Development of Cross-correlation Function

The derivation begins by assuming the standard matrix equations of motion:

$$[M]\,\{\ddot{x}(t)\} + \{\dot{x}(t)\} + [K]\,\{x(t)\} = \{f(t)\} \tag{1}$$

where

$[M]$ is the mass matrix
$[C]$ is the damping matrix

$[K]$ is the stiffness matrix
$\{f\}$ is a vector of random forcing functions
$\{x\}$ is the vector of random displacements.

Equation (1) can be expressed in modal coordinates using a standard modal transformation:

$$\{x(t)\} = [\Psi]\{q(t)\} = \sum_{r=1}^{n} \{\Psi_r\} q^r(t) \tag{2}$$

where
$[\Psi]$ is the modal matrix;
$\{q(t)\}$ is a vector of modal coordinates;
$\{\Psi_r\}$ is the rth mode shape; and
n is the number of modes.

A premultiplication of Eq. (1) by $[\Psi]^T$ is also performed. Since real normal modes are assumed, $[M]$, $[C]$, and $[K]$ are diagonalized. The result is a set of scalar equations in the modal coordinates:

$$\ddot{q}^r(t) + 2\zeta^r \omega_n^r \dot{q}^r(t) + \omega_n^{r^2} q^r(t) = \frac{1}{m^r} \{\Psi_r\}^T \{f(t)\} \tag{3}$$

where
ω_n^r is the rth modal frequency
ζ^r is the rth modal damping ratio
m^r is the rth modal mass.

The solution of Eq. (3), assuming a general $\{f\}$ and zero initial conditions, is obtained from the convolution or Duhamel integral [27]:

$$q^r(t) = \int_{-\infty}^{t} \{\Psi_r\}^T \{f(\tau)\} g^r(t - \tau) d\tau \tag{4}$$

where $g^r(t) = 0$, for $t < 0$; $g^r(t) = (1/m^r\omega_d^r)\exp(-\zeta^r\omega_n^r t)\sin(\omega_n^r t)$, for $t \geq 0$; and $\omega_d^r = \omega_n^r(1 - \zeta^{r^2})^{1/2}$ is the damped modal frequency.

Equations (4) and (2) can now be used to obtain the solution for $\{x(t)\}$:

$$\{x(t)\} = \sum_{r=1}^{n} \{\Psi_r\} \cdot \int_{-\infty}^{t} \{\Psi_r\}^T \{f(\tau)\} g^r(t - \tau) d\tau \tag{5}$$

where n is the number of modes.

Equation (5) will now be specialized for a single output, $x_{ik}(t)$, due to a single input force, $f_k(t)$, at point k.

$$x_{ik}(t) = \sum_{r=1}^{n} \Psi_{ir} \Psi_{kr} \cdot \int_{-\infty}^{t} f_k(\tau) g^r(t - \tau) d\tau \tag{6}$$

where Ψ_{ir} is the ith component of mode shape r.

The impulse response function between input k and output i results when $f(\tau)$ in Eq. (6) is a Dirac delta function at $\tau = 0$. The integration collapses with the following results:

$$x_{ik}(t) = \sum_{r=1}^{n} \frac{\Psi_{ir}\,\Psi_{kr}}{m^r \omega_d^r} \exp(-\zeta^r \omega_n^r t)\sin(\omega_d^r t). \tag{7}$$

The next step of the theoretical development is to form the cross-correlation function of two responses (x_{ik} and x_{jk}) due to a white-noise input at a particular input point k. Reference [28] defines the cross-correlation function $\Re_{ijk}(T)$ as the expected value of the product of two responses evaluated at a time separation of T:

$$\Re_{ijk}(T) = E[x_{ik}(t + T)\, x_{jk}(t)] \tag{8}$$

where E is the expectation operator.

Substituting Eq. (6) into (8) results in the following, since $f_k(t)$ is the only random variable:

$$\Re_{ijk}(T) = \sum_{r=1}^{n} \sum_{s=1}^{n} \Psi_{ir}\,\Psi_{kr}\,\Psi_{js}\,\Psi_{ks}\cdot \int_{-\infty}^{t} \int_{-\infty}^{t+T} g^r(t + T - \sigma)\, g^s(t - \tau)\, E[f_k(\sigma) f_k(\tau)] d\sigma\, d\tau. \tag{9}$$

Using the definition of the autocorrelation function [28], and assuming $f(t)$ of Eq. (9) is white noise, then the autocorrelation function of f is:

$$\Re_{ffk}(\tau - \sigma) = E[f_k(\tau) f_k(\sigma)] = \alpha_k \delta(\tau - \sigma) \tag{10}$$

where α_k is a constant and $\delta(t)$ is the Dirac delta function.

Substituting Eq. (10) into Eq. (9), and collapsing the first integration by using the definition of the delta function produces the following:

$$\Re_{ijk}(T) = \sum_{r=1}^{n} \sum_{s=1}^{n} \alpha_k\,\Psi_{ir}\,\Psi_{kr}\,\Psi_{js}\,\Psi_{ks}\cdot \int_{-\infty}^{t} g^r(t + T - \tau)\, g^s(t - \tau) d\tau. \tag{11}$$

Equation (11) can be further simplified by making a change in the variable of integration. If $\lambda = t - \tau$, then the limits of integration are zero and ∞. And Eq. (11) becomes:

$$\Re_{ijk}(T) = \sum_{r=1}^{n} \sum_{s=1}^{n} \alpha_k\,\Psi_{ir}\,\Psi_{kr}\,\Psi_{js}\,\Psi_{ks}\cdot \int_{0}^{\infty} g^r(\lambda + T)\, g^s(\lambda) d\lambda. \tag{12}$$

Using the definition of g from Eq. (4) and the trig identity for the sine of a sum results in all the terms involving T separating from those involving λ:

$$\begin{aligned} g^r(\lambda + T) = {} & [\exp(-\zeta^r \omega_n^r T) \cos(\omega_d^r T)] \frac{\exp(-\zeta^r \omega_n^r \lambda) \sin(\omega_d^r \lambda)}{m^r \omega_d^r} \\ & + [\exp(-\zeta^r \omega_n^r T) \sin(\omega_d^r T)] \frac{\exp(-\zeta^r \omega_n^r \lambda) \cos(\omega_d^r \lambda)}{m^r \omega_d^r}. \end{aligned} \tag{13}$$

Note that substitution of Eq. (13) into (12) along with the corresponding formula for $g^s(\lambda)$ allows terms that depend on T to be factored out of the integral and out of the second summation (the s index), resulting in:

$$\Re_{ijk}(T) = \sum_{r=1}^{n} [A^r_{ijk} \exp(-\zeta^r\omega^r_n T) \cos(\omega^r_d T) + B^r_{ijk} \exp(-\zeta^r\omega^r_n T) \sin(\omega^r_d T)] \tag{14}$$

where A^r_{ijk} and B^r_{ijk} are independent of T, are functions of only the modal parameters, contain completely the summation on s, and are shown below.

$$\begin{Bmatrix} A^r_{ijk} \\ B^r_{ijk} \end{Bmatrix} = \sum_{s=1}^{n} \frac{\alpha_k\, \Psi_{ir}\, \Psi_{kr}\, \Psi_{js}\, \Psi_{ks}}{m^r\omega^r_d m^s \omega^s_d} \cdot \int_0^\infty \exp(-\zeta^r\omega^r_n - \zeta^s\omega^s_n)\lambda \cdot \sin(\omega^s_d\lambda) \begin{Bmatrix} \sin(\omega^r_d\lambda) \\ \cos(\omega^r_d\lambda) \end{Bmatrix} d\lambda. \tag{15}$$

Equation (14) is the key result of this derivation. Examining Eq. (14), the cross-correlation function is seen to be a sum of decaying sinusoids, with the same characteristics as the impulse response function of the original system (see Eq. (7)); thus, cross-correlation functions can be used as impulse response functions in time-domain modal parameter estimation schemes.

5. Simplification of Cross-correlation Function

A^r_{ijk} and B^r_{ijk} can be further simplified by evaluating the definite integral:

$$A^r_{ijk} = \sum_{s=1}^{n} \frac{\alpha_k\, \Psi_{ir}\, \Psi_{kr}\, \Psi_{js}\, \Psi_{ks}}{m^r m^s \omega^r_d} \left[\frac{I_{rs}}{J^2_{rs} + I^2_{rs}}\right] \tag{16}$$

$$B^r_{ijk} = \sum_{s=1}^{n} \frac{\alpha_k\, \Psi_{ir}\, \Psi_{kr}\, \Psi_{js}\, \Psi_{ks}}{m^r m^s \omega^r_d} \left[\frac{J_{rs}}{J^2_{rs} + I^2_{rs}}\right] \tag{17}$$

where $I_{rs} = 2\omega^r_d(\zeta^r\omega^r_n + \zeta^s\omega^s_n)$ and $J_{rs} = (\omega^{s^2}_d - \omega^{r^2}_d) + (\zeta^r\omega^r_n + \zeta^s\omega^s_n)^2$.

To further illustrate the useful form of these results, define a quantity γ_{rs} such that:

$$\tan(\gamma_{rs}) = I_{rs}/J_{rs}. \tag{18}$$

Using this relationship in Eq. (16) and (17) provides:

$$A^r_{ijk} = \frac{\Psi_{ir}}{m^r\omega^r_d} \sum_{s=1}^{n} \beta^{rs}_{jk}(J^2_{rs} + I^2_{rs})^{-1/2} \sin(\gamma_{rs}) \tag{19}$$

and

$$B^r_{ijk} = \frac{\Psi_{ir}}{m^r\omega^r_d} \sum_{s=1}^{n} \beta^{rs}_{jk}(J^2_{rs} + I^2_{rs})^{-1/2} \cos(\gamma_{rs})$$

where $\beta^{rs}_{jk} = \alpha_k\, \Psi_{kr}\, \Psi_{js}\, \Psi_{ks}/m^s$.

Substituting Eq. (19) into Eq. (14), and summing over all the input locations, m, to find the cross-correlation function due to all the inputs (denoted by R_{ij}):

$$R_{ij}(T) = \sum_{r=1}^{n} \frac{\Psi_{ir}}{m^r \omega_d^r} \sum_{s=1}^{n} \sum_{k=1}^{n} \beta_{jk}^{rs} (J_{rs}^2 + I_{rs}^2)^{-1/2} \exp(-\zeta^r \omega_n^r T) \sin(\omega_d^r T + \gamma_{rs}). \tag{20}$$

The inner summations on s and k are merely a summation of constants times the sine function, with variable phase but fixed frequency. Equation (20) can therefore be rewritten as a single sine function with a new phase angle (ϑ^r) and a new constant multiplier (G_j^r):

$$R_{ij}(T) = \sum_{r=1}^{n} \frac{\Psi_{ir} G_j^r}{m^r \omega_d^r} \exp(-\zeta^r \omega_n^r T) \sin(\omega_d^r T + \vartheta^r). \tag{21}$$

This completes the theoretical development for the single input, multi-output, multi-mode case. It shows that the cross-correlation function in Eq. (21) is a sum of decaying sinusoids of the same form as the impulse response function of the original system in Eq. (7).

6. Verification Using Cross-correlation Identity

A check of the preceeding development is to verify the following identity [26]:

$$R_{ij}(T) = R_{ji}(-T). \tag{22}$$

Equation (12) will be explored for the case of $T < 0$. First a new positive variable is defined:

$$T = -T_n \tag{23}$$

Then a new variable of integration is defined:

$$\rho = \lambda - T_n. \tag{24}$$

The following relationships then result:

$$\lambda = \rho + T_n; \tag{25}$$

$$d\lambda = d\rho; \tag{26}$$

$$\text{at } \lambda = 0,\ \rho = -T_n; \text{ and} \tag{27}$$

$$\text{at } \lambda = \infty,\ \rho = \infty \tag{28}$$

Upon substitutingn Eqs. (23–28) into (12) the following results:

$$\Re_{ijk}(-T_n) = \sum_{r=1}^{n} \sum_{s=1}^{n} \alpha_k \, \Psi_{ir} \, \Psi_{kr} \, \Psi_{js} \, \Psi_{ks} \int_{-T_n}^{\infty} g^r(\rho) g^s(\rho + T_n) d\rho. \tag{29}$$

Now the integration interval can be split:

$$\Re_{ijk}(-T_n) = \sum_{r=1}^{n} \sum_{s=1}^{n} \alpha_k\, \Psi_{ir}\, \Psi_{kr}\, \Psi_{js}\, \Psi_{ks} \cdot \int_0^{\infty} g^r(\rho) g^s(\rho + T_n) d\rho$$

$$+ \sum_{r=1}^{n} \sum_{s=1}^{n} \alpha_k\, \Psi_{ir}\, \Psi_{kr}\, \Psi_{js}\, \Psi_{ks} \cdot \int_{-T_n}^{0} g^r(\rho) g^s(\rho + T_n) d\rho. \tag{30}$$

From the Eq. (4), $g^r(\rho) = 0$ for ρ between $-T_n$ and 0 which drives the second term in Eq. (30) to zero. Also the first term can be rewritten by interchanging the r and s variables:

$$\Re_{ijk}(-T_n) = \sum_{s=1}^{n} \sum_{r=1}^{n} \alpha_k\, \Psi_{ir}\, \Psi_{kr}\, \Psi_{js}\, \Psi_{ks} \cdot \int_0^{\infty} g^s(\rho + T_n) g^r(\rho) d\rho + 0$$

$$= \sum_{r=1}^{n} \sum_{s=1}^{n} \alpha_k\, \Psi_{ir}\, \Psi_{kr}\, \Psi_{js}\, \Psi_{ks} \cdot \int_0^{\infty} g^r(\zeta + T_n) g^s(\zeta) d\rho = \Re_{jik}(T_n). \tag{31}$$

Therefore, the identity given in Eq. (22) has been proven after summing over k and substituting Eq. (23) for T_n:

$$R_{ij}(T) = R_{ji}(-T). \tag{22}$$

7. Verification Using Autocorrelation Maximum

A further check of the validity of theoretical development is to show that the autocorrelation function is a maximum when T is zero in Eq. (14). The verification is initiated by taking the derivative of the cross-correlation function as provided by Eq. (14):

$$\frac{d}{dT}[\Re_{ijk}(T) = \frac{d}{dT}\left[\sum_{r=1}^{n} A_{ijk}^r \exp(-\zeta^r \omega_n^r T) \cos(\omega_d^r T) \sin(\omega_d^r T)\right.$$

$$\left. + B_{ijk}^r \exp(-\zeta^r \omega_n^r T) \sin(\omega_d^r T)\right]$$

$$= \sum_{r=1}^{n} \{[\omega_d^r B_{ijk}^r - \zeta^r \omega_n^r A_{ijk}^r] \exp(-\zeta^r \omega_n^r T) \cos(\omega_d^r T) \tag{32}$$

$$- [\omega_d^r A_{ijk}^r + \zeta^r \omega_n^r B_{ijk}^r] \exp(-\zeta^r \omega_n^r T) \cos(\omega_d^r T)\}$$

$$= 0, \text{ for a maximum.}$$

Now, at $T = 0$ the following results:

$$\frac{d}{dT}[\Re_{ijk}(0)] = \sum_{r=1}^{n} (\omega_d^r B_{ijk}^r - \zeta^r \omega_n^r A_{ijk}^r) = 0, \text{ for a maximum.} \tag{33}$$

Substituting Eqs. (15) into (33) and rearranging provides the following:

$$\frac{d}{dT}[\Re_{ijk}(0)] = \sum_{r=1}^{n}\sum_{s=1}^{n} \frac{\alpha_k \Psi_{ir} \Psi_{kr} \Psi_{js} \Psi_{ks}}{m^r \omega_d^r m^s \omega_d^s}$$

$$\cdot \left\{ \omega_d^r \int_0^\infty \exp[-(\zeta^r \omega_n^r + \zeta^s \omega_n^s)\lambda] \sin(\omega_d^s \lambda) \cos(\omega_d^r \lambda) d\lambda \right.$$

$$\left. - \zeta^r \omega_n^r \int_0^\infty \exp[-(\zeta^r \omega_n^r + \zeta^s \omega_n^s)\lambda] \sin(\omega_d^s \lambda) \sin(\omega_d^r \lambda) d\lambda \right\}$$

$$= 0, \text{ for a maximum.} \tag{34}$$

Now performing the integrations and collecting terms results in the following:

$$\frac{d}{dT}[\Re_{ijk}(0)] = \sum_{r=1}^{n}\sum_{s=1}^{n} \frac{\alpha_k \Psi_{ir} \Psi_{kr} \Psi_{js} \Psi_{ks} N^{rs}}{D^{rs}}$$

$$\text{where } N^{rs} = (\omega_n^{s^2} - \omega_n^{r^2}), \text{ and}$$

$$D^{rs} = m^r m^s [(\zeta^s \omega_n^s + \zeta^r \omega_n^r)^2 + (\omega_n^s + \omega_n^r)^2]$$

$$\cdot [(\zeta^s \omega_n^s + \zeta^r \omega_n^r)^2 + (\omega_n^s + \omega_n^r)^2]. \tag{35}$$

The following identities are seen to hold by inspection:

$$N^{rs} = 0, \text{ when } r = s. \tag{36}$$

$$N^{rs} = -N^{sr}, \text{ when } r \neq s. \tag{37}$$

$$D^{rs} = D^{sr}. \tag{38}$$

Hence Eq. (35) sums to zero when $i = j$. Therefore, an autocorrelation function described by Eq. (14) reaches a maximum at $T = 0$. This provides another validation of the development leading to Eq. (14). The next sections further verify NExT using simulated and experimental data.

8. Verification of NExT Using Simulated Data

A simulation code, VAWT-SDS, has been produced to compute the time domain response of VAWTs during rotation in turbulent wind [29]. The structural model used in VAWT-SDS was available to calculate the analytical modal frequencies and analytical modal damping. VAWT-SDS was used to generate analytical data, which were then input to NExT. The results were compared to the known frequencies and damping information to test the capabilities of NExT.

The specific machine modeled was the DOE/Sandia 34-m testbed located in Bushland, Texas. Figure 1 shows a schematic of this turbine. The blades and supporting tower are rotating components designed to convert wind energy into electricity. Simulated data were generated for the 34-m testbed using a 30 rpm rotation rate and 20 mph turbulent winds with a 15% turbulence intensity. Stiffness

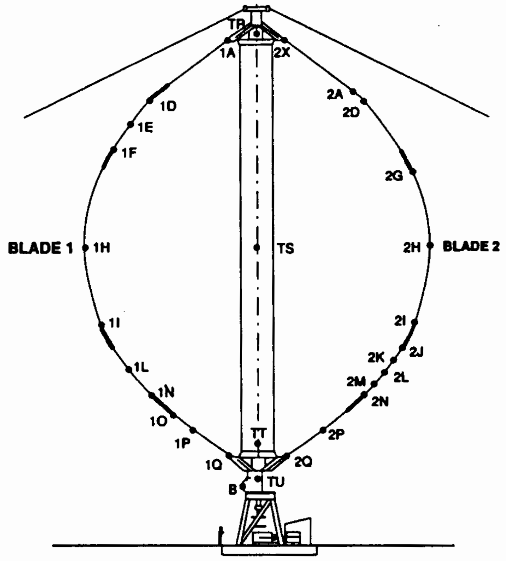

Fig. 1 Schematic of DOE/Sandia 34-m testbed with strain gage locations marked

proportional damping, sufficient to produce a damping ratio of 0.2% at 1.4 Hz was added to the model. Time histories of 2048 points for eight strain gauge outputs were generated using a step size of 0.04 s. Ten sets of these time histories were calculated with similar wind conditions to allow ensemble averaging. Sensor noise was simulated by adding a white-noise signal to each simulated time history. The standard deviation of this additive signal was 2% of the standard deviation of each time history.

The analytical modal frequencies and damping ratios were calculated by extracting the complex eigenvalues from the structural matrices used in the VAWT-SDS code. VAWT-SDS used the Newmark-Beta numerical integration scheme, and the approximations inherent in this procedure produced period elongations [30]. The frequency shifts created by numerical integration were calculated and a correction was added to the analytical values [15]. NExT was then used to estimate modal frequencies and damping ratios from the simulated data so that the NExT results and the analytical values could be compared. Polyreference was used to perform the time-domain analysis.

Figures 2 and 3 are examples of parked VAWT mode shapes. These shapes will be referred to throughout this report. Table 1 shows the results of the comparison. The agreement between the actual and the NExT generated modal frequencies is excellent. The only exceptions to this are the first two modes at 1.27 Hz and 1.35 Hz. These modes are very closely spaced, making it difficult to obtain accurate results from NExT. Generally, the agreement between the VAWT-SDS specified damping ratios and the calculated damping ratios is good, with a few exceptions. The damping ratios for the first flatwise antisymmetric (1.27 Hz) and the first flatwise symmetric (1.35 Hz) modes were not estimated well because these modes could not be separated. The higher modes (3.65 Hz, 3.73 Hz, and 3.88 Hz) have NExT estimated damping ratios that are lower than the specified damping ratios. The amplitudes of these modes are low compared to the noise level, which adversely affected the estimates. Obtaining longer time histories would improve the accuracy. However, considering the precision of experimental damping ratio estimates, these values are still quite acceptable. The ability of NExT to reproduce known modal frequencies and specified damping levels lends confidence for its application to field data.

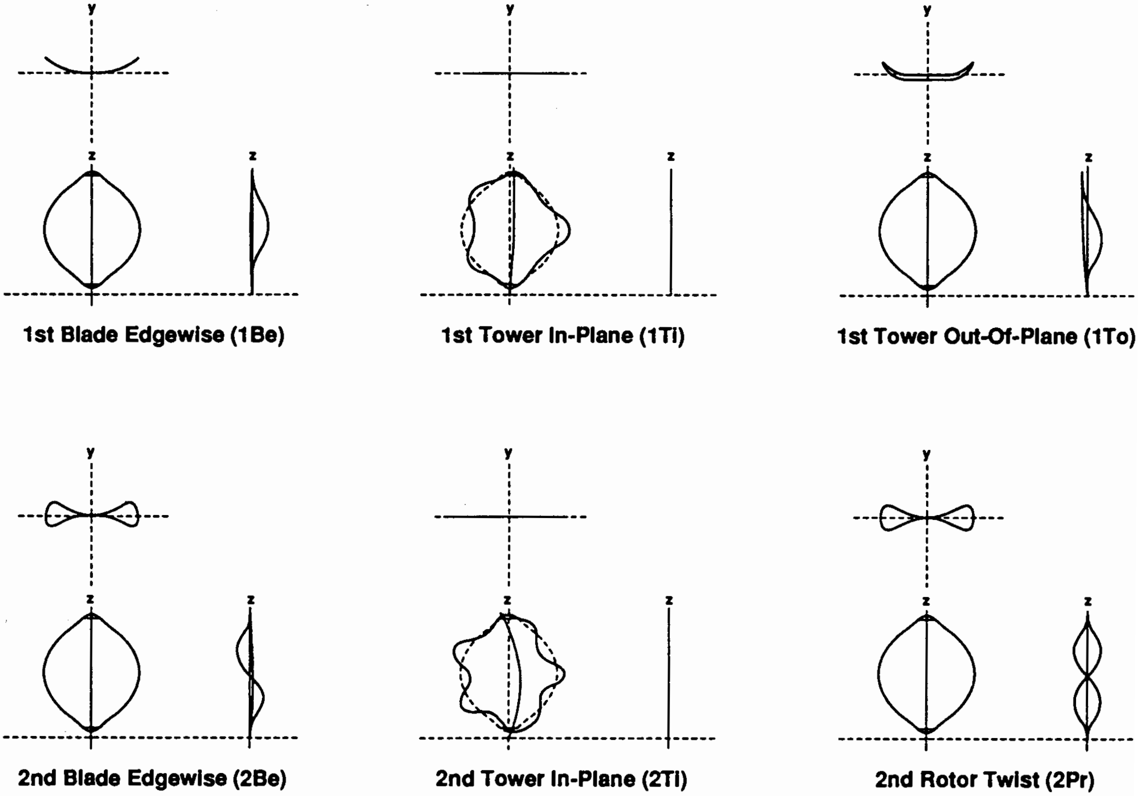

Fig. 2 Parked VAWT non-flatwise mode shapes

9. Comparison of NExT and Conventional Modal Test Results

A FloWind Corporation 19-m VAWT in Altmont Pass, CA, was tested using conventional modal testing techniques [31] during quiescent daytime winds. NExT was then used during periods of more substantial nighttime winds (above 7 m/s or 16 mph). The turbine was parked (nonrotating) during all testing. Accelerometers were used to measure the response at predetermined locations on the turbine. This allowed a comparison between modal parameters estimated by NExT and modal parameters estimated using more conventional techniques.

Table 2 compares the modal frequencies and modal damping ratios of the 19-m VAWT as determined from the conventional step relaxation testing and from NExT. The two methods produced estimates of the modal frequencies that are in good agreement, particularly in view of the temperature difference between day and night. The average difference for the ten modes is only 0.5%. Also, the modal damping ratios of all six of the tower modes (rotor twist, tower in-plane, tower out-of-plane) are very similar. However, the modal damping ratios of all four blade flatwise modes (flatwise symmetric and flatwise antisymmetric) are substantially higher from NExT estimates. This difference is believed to be due to a drag phenomenon similar to that experienced by an oscillating flat-plate normal to a strong wind [32, 33]. Reference [18] contains the results of a scoping calculation, which predicts 1.2% damping from aero drag effects for the first blade flatwise modes. This would account for the 1.1% and 1.3% additional damping for the first blade flatwise modes as seen in Table 2. The utility of performing modal testing in an operating environment is shown by this phenomena. Since the traditional modal testing was performed in quiescent winds, this additional damping mechanism was

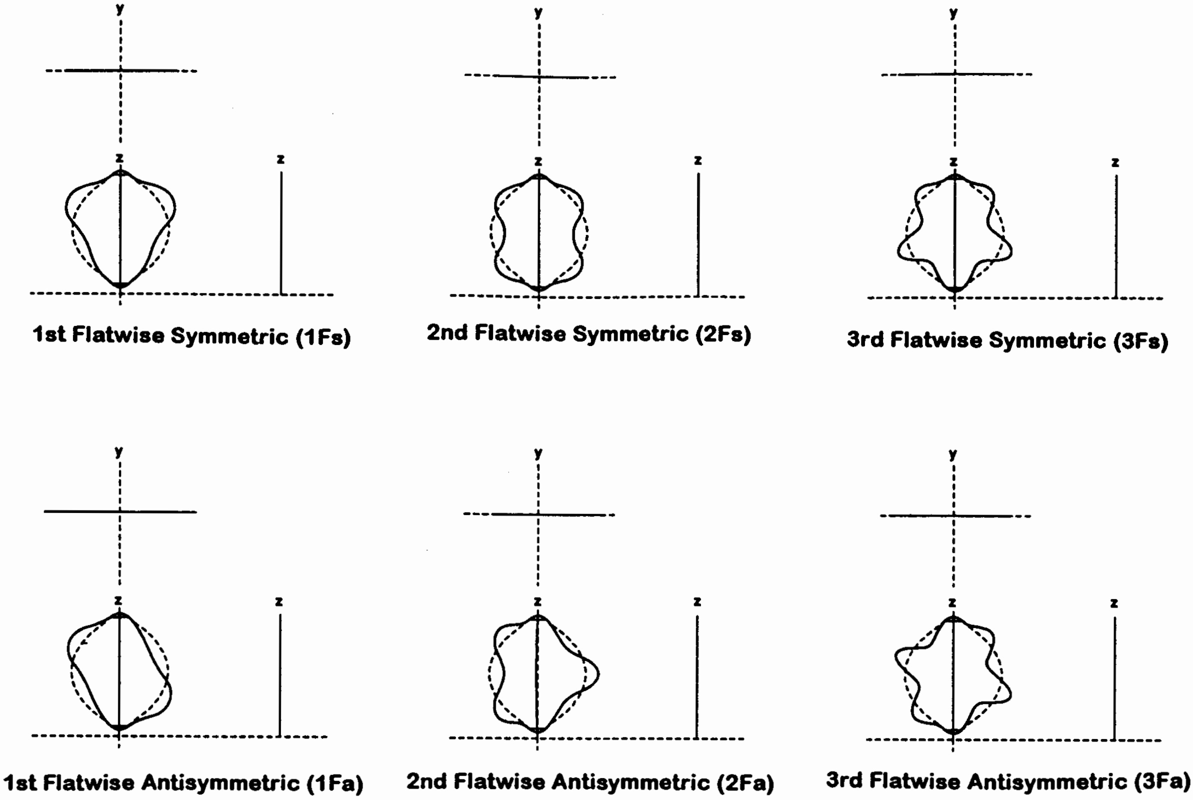

Fig. 3 Parked VAWT flatwise mode shapes

TABLE 1 COMPARISON OF NExT WITH SIMULATED RESULTS

	Frequency (Hz)		Damping (%)	
Mode	Simulated	NExT	Simulated	NExT
1st Flatwise Antisymmetric	1.27	1.31	0.2	0.4
1st Flatwise Symmetric	1.35	1.32	0.2	0.3
1st Blade Edgewise	1.59	1.59	0.3	0.3
1st Tower In-Plane	2.02	2.01	0.3	0.4
2nd Flatwise Symmetric	2.43	2.44	0.4	0.5
2nd Flatwise Antisymmetric	2.50	2.50	0.4	0.4
1st Tower Out-of-Plane	2.80	2.80	0.3	0.5
2nd Rotor Twist	3.39	3.39	0.5	0.6
2nd Tower In-Plane	3.46	3.45	0.5	0.4
3rd Flatwise Antisymmetric	3.65	3.63	0.5	0.4
3rd Flatwise Symmetric	3.73	3.73	0.6	0.4
2nd Blade Edgewise	3.88	3.87	0.5	0.3

only active during natural excitation testing. Polyreference was again used to perform the time-domain analysis. Overall, the comparison of results from NExT with those obtained using conventional methods further verifies the accuracy of NExT.

The presence of the air drag phenomenon shows that NExT extracts the total damping, structural and aeroelastic. This is the desired situation for operational testing because aeroelastic damping will be added to structural damping during rotation. However, for parked turbine testing, which is designed

TABLE 2 COMPARISON OF NExT WITH EXPERIMENTAL RESULTS

Mode	Frequency (Hz) Step Relax	Frequency (Hz) NExT	Damping (%) Step Relax	Damping (%) NExT
1st Rotor Twist	2.37	2.38	0.2	0.1
1st Flatwise Antisymmetric	2.48	2.49	0.2	1.3
1st Flatwise Symmetric	2.51	2.51	0.1	1.4
1st Tower Out-of-Plane	2.72	2.76	0.4	0.4
1st Tower In-Plane	3.11	3.15	0.4	0.4
2nd Tower Out-of-Plane	4.53	4.53	0.1	0.1
2nd Flatwise Antisymmetric	5.30	5.31	0.3	0.8
2nd Flatwise Symmetric	5.64	5.65	0.1	0.6
2nd Rotor Twist	6.59	6.62	0.1	0.1
2nd Tower In-Plane	6.64	6.71	0.3	0.6

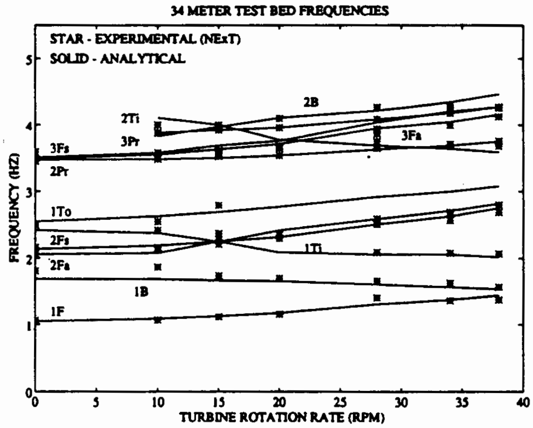

Fig. 4 Modal frequencies as a function of turbine rotation rate

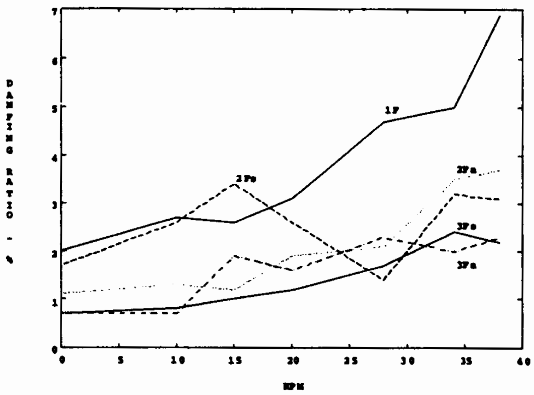

Fig. 5 Damping of flatwise modes

to capture structural damping, conventional modal testing techniques may be more appropriate. The need for quiescent winds during such testing is even more apparent from these results.

10. NExT Results From a Rotating VAWT

VAWTs undergo significant aeroelastic and rotational loads and require validated structural models (and therefore modal testing) for various operating conditions. Specific interest is placed on determining the modal damping that arises from aeroelastic interactions. NExT has been used to extract modal damping using data from the DOE/Sandia 34-m testbed during rotation at 0, 10, 15, 20, 28, 34, and 38 rpm. The wind speed during these tests was ~10 m/s, or 22 mph. Twelve strain gauges were used as the sensors. Time histories of 30 minutes were recorded at 20 samples per second. These time histories were used to calculate averaged correlation functions of 1024 points. ERA was used to perform the time-domain analysis. Figure 4 is a plot of the modal frequencies of the 34-m testbed from analysis and from experiment using NExT. This plot shows the analytical modal frequencies changing as a function of rotation rate and the associated frequencies measured using NExT.

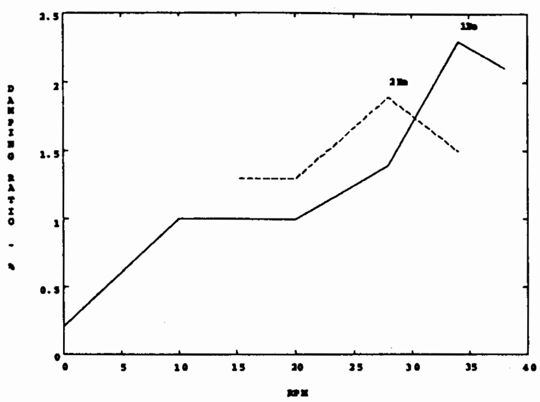

Fig. 6 Damping of first and second blade edgewise modes

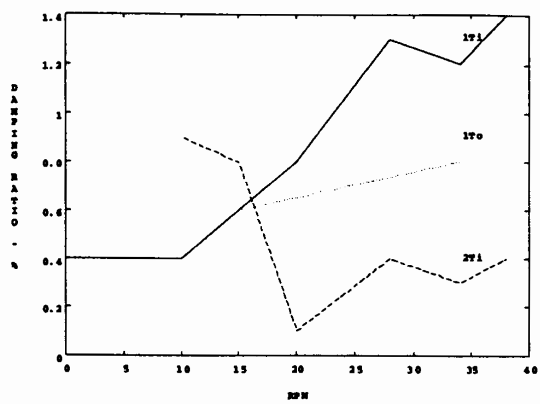

Fig. 7 Damping of tower in-plane and tower out-of-plane modes

11. Damping Versus Wind Turbine Rotation Rate

Figure 5 shows a plot of modal damping ratio, as calculated with NExT, versus turbine rotation rate for the blade flatwise modes. The damping ratios of these modes generally increase with turbine rotation rate. The increase in damping is quite significant; for example, the first flatwise damping increases from 2% to 7%. The notable exception is the second blade flatwise mode between 15 rpm and 28 rpm. Such a drop in damping ratio could be due to modal coupling to a more lightly damped mode, since the modal coupling varies with rotation speed. The resulting mode shape could be less affected by the aerodynamic damping terms and, therefore, the damping ratio would drop. Mode shape information is needed to answer this question; however, shape information has not been extracted from this data set.

Figure 6 shows a plot of modal damping ratio versus turbine ratio for the blade edgewise modes. The trend of increasing damping ratio as rotation rate increases is again seen. The magnitudes of these modal damping ratios are significantly smaller than those of the flatwise modes. Figure 7 shows a plot of the same data for the tower in-plane and tower out-of-plane modes. The damping of the first tower modes have an increasing trend, as seen before. The second tower in-plane mode has an unexpected drop in damping ratio at 20 rpm.

Figure 8 contains information about the second and third propeller or rotor twist modes. The notable feature of this plot is the large change in modal damping of the second propeller mode at 10 rpm. The change could be due to coupling with a more highly damped mode. Again, mode shape information will help to clear up this question.

12. Autospectrum Synthesis

Figure 9 shows actual test data overlaid with a synthesis of the autospectrum of lead-lag strain at the bottom of a 34-m testbed blade while operating at 28 rpm. This figure illustrates the usefulness and the limits of this graphical check as well as several aspects of the modal parameter estimation process. The labels 1P, 2P, 3P, 4P, and 5P denote the per-rev harmonics. These harmonics are seen to be captured well, although NExT currently fits the harmonics as though they were actual modes.

The first tower in-plane mode, 1Ti, and the second propeller mode (rotor twist), 2Pr, are a good fit. The near exact synthesis of the position and slope of the peak gives confidence in the modal

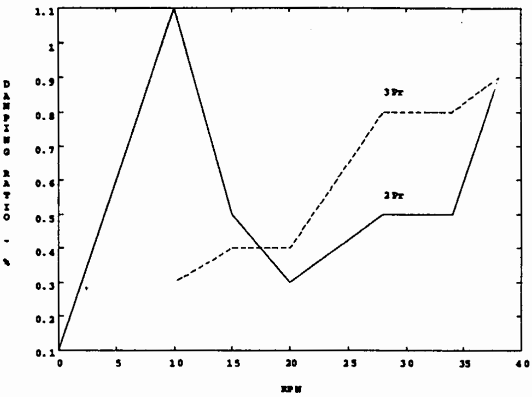

Fig. 8 Damping of second and third rotor twist modes

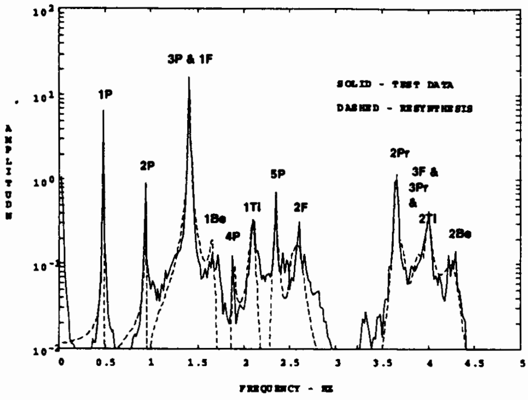

Fig. 9 Autospectrum and synthesis of lead-lag strain at 28 rpm

frequency and damping ratio from NExT as well as the amplitude coefficient calculated by the autospectrum synthesis. The first blade edgewise mode, 1Be, and the second blade flatwise modes, 2F, are seen to have the correct modal frequency estimates, but the autospectrum coefficient was not estimated closely. This could be due to an incomplete mathematical representation of the autospectrum, noise in the experimental autospectrum, or numerical interactions with adjacent modes. The autospectrum synthesis provides little information about the quality of the modal damping ratio estimates in such cases.

The first blade flatwise modes, 1F, are coincident with the 3P harmonic. The autospectrum synthesis provides no information about the quality of the modal damping ratio estimate and very little information about the modal frequency estimate in such situations. A technique to remove the 3P harmonic would probably improve the results. The third blade flatwise modes (3F), the third propeller mode (3Pr), and the second tower in-plane mode (2Ti) are closely grouped. Mode shape information is needed to separate these modes. Finally, the second blade edgewise (2Be) is in a region of low signal-to-noise ratio. Information on the fit of this mode should be obtained from different data channels.

The comparison of the NExT synthesised autospectrum to the autospectrum of actual operating data shows excellent agreement in this typical example. However, a few suggestions for improvement can be made. The technique could be improved by providing a method for extracting shape information that could alleviate some problems associated with identifying modes, separating closely spaced modes, or assessing the degree of coupling induced by dynamic conditions. Advanced modal analysis techniques [34] are available which could enhance the low amplitude modes, making it easier to estimate their modal parameters.

13. Comparison of NExT Results With Analysis for a Rotating VAWT

There are currently two reports that contain analytical predictions of the aeroelastic damping expected during operation of the Sandia/DOE 34-m testbed. Reference 20 reports the work of Lobitz and Ashwill in which the NASTRAN generated damping and stiffness matrices are modified to include the effects of aeroelasticity. Reference 21 includes the work by Malcolm using a modified form of the Lobitz and Ashwill aeroelasticity model. Malcolm's modifications included terms that model a rotating coordinate frame and an elastic center offset. These references provide the specific information on the assumed analytical models.

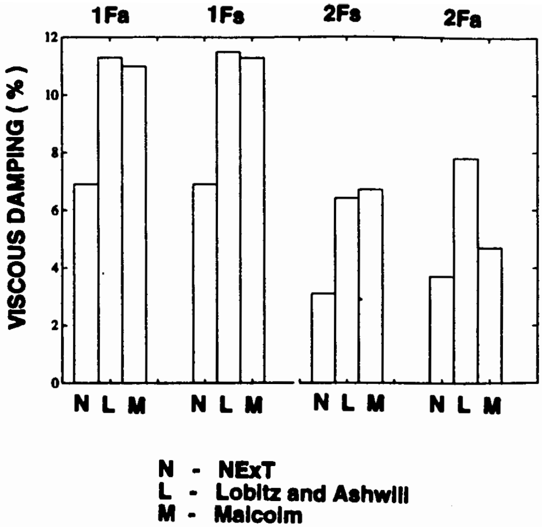

Fig. 10 Analytical damping results for flatwise modes at 38 rpm

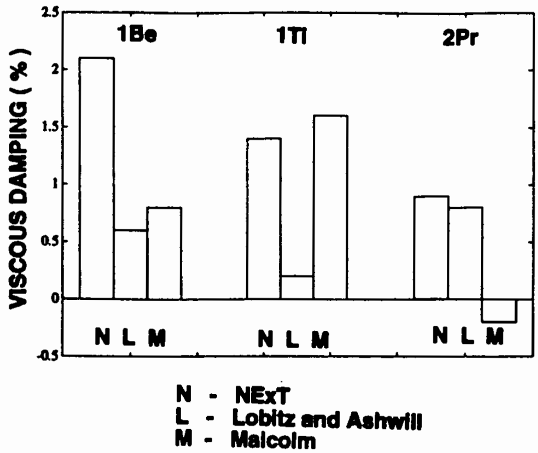

Fig. 11 Analytical damping results for non-flatwise modes at 38 rpm

Figures 10 and 11 compare these analytical results with the damping calculations of NExT. The turbine rotation rate was between 38 and 40 rpm for the data in these comparisons. Figure 10 compares the results for four flatwise modes. The reader is reminded that NExT could not separate the first flatwise modes. However, a clear trend still results, with the predicted damping being higher than the NExT estimates. The damping values are shown as percent of critical viscous damping. Figure 11 compares results for three non-flatwise modes. No clear trend can be seen in these results.

14. Summary

A brief overview of NExT has been presented, followed by a theoretical justification. Analytically generated data have been used to verify the ability of NExT to extract modal frequencies and damping ratios from operating data. NExT was further verified by a comparison with conventional modal testing techniques using a parked vertical-axis wind turbine. This same comparison shows the ability of NExT to estimate the total damping of the system. An extra damping mechanism for the flatwise modes of a parked VAWT was implied. This is due to the drag force experienced by a flat plate oscillating parallel to a flowing fluid.

Damping versus turbine rotation rate plots have been presented for the DOE/Sandia 34-m testbed over a range of operational rates. A general trend was shown of substantially increasing damping as rotation rate increases. Comparing a synthesized autospectrum with a test data autospectrum is useful for verifying the estimates of the modal parameters. Mode shape information is needed to help explain sudden changes in damping ratio as turbine rotation rate increases, to aid in the identification of higher frequency modes and for verification of autospectrum synthesis coefficients. Advanced modal analysis techniques for extracting information from low-amplitude modes and filtering techniques to remove harmonic peaks are desirable upgrades to NExT.

Comparisons between NExT generated results and analytical predictions based on aeroelastic theory illustrate how NExT can be used to provide the necessary information to refine the analytical predictions. The predicted damping of the blade flatwise modes was higher than the damping estimates provided by NExT. However, no clear difference was seen in the predicted damping for the non-flatwise modes. The inability of NExT to separate the first blade symmetric and antisymmetric modes indicates the need to develop techniques to separate closely spaced modes.

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